

VELLORE INSTITUTE OF TECHNOLOGY

School of Computer Science and Engineering

BCSE302L - DATABASE SYSTEMS (LAB COURSE PROJECT)

COLLEGE MANAGEMENT SYSTEM

A Normalized, High-Integrity Relational Database System

REVIEW I REPORT: Problem Definition, ER Modeling & Relational Schema

FACULTY IN-CHARGE:

Prof. RAMKUMAR T

School of Computer Science and Engineering

STUDENT DEVELOPMENT TEAM

- **ANSH** **Reg No: 24BDE0019**
B.Tech Data Engineering | Role: Lead Architect & Schema Specialist
- **ANEEL CHATTERJEE** **Reg No: 24BYB0093**
B.Tech Computer Science | Role: PL/SQL Backend & Stored Logic Developer
- **PARBHAT RANJAN** **Reg No: 24BDE0086**
B.Tech Data Engineering | Role: Analytical Queries & Transaction Automation

1. PROBLEM DEFINITION & SCOPE

BCSE302L - REVIEW I

2. 8-TABLE RELATIONAL SCHEMA & CONSTRAINTS

BACHELOR - REVIEW I

1. DEPARTMENT

dept_id (PK), dept_name (UQ, NN), dept_code (UQ, NN), building (NN), budget (CHECK > 0), established_year (CHECK 1950..2020)

2. FACULTY

faculty_id (PK), first_name (NN), last_name (NN), email (UQ, NN), phone (UQ, NN), hire_date (NN), designation (CHECK), salary

3. STUDENT

student_id (PK), reg_no (UQ, NN), first_name (NN), last_name (NN), email (UQ, NN), phone (UQ, NN), dob (NN), gender (CHECK)

4. COURSE

course_id (PK), course_code (UQ, NN), course_title (NN), credits (CHECK 1..6), dept_id (FK -> DEPARTMENT), course_level (CHECK)

5. COURSE_OFFERING

offering_id (PK), course_id (FK -> COURSE), faculty_id (FK -> FACULTY), academic_year (NN), semester (CHECK 1..8), classroom

6. ENROLLMENT

enrollment_id (PK), student_id (FK -> STUDENT), offering_id (FK -> COURSE_OFFERING), enrollment_date (DEFAULT), status (CHECK)

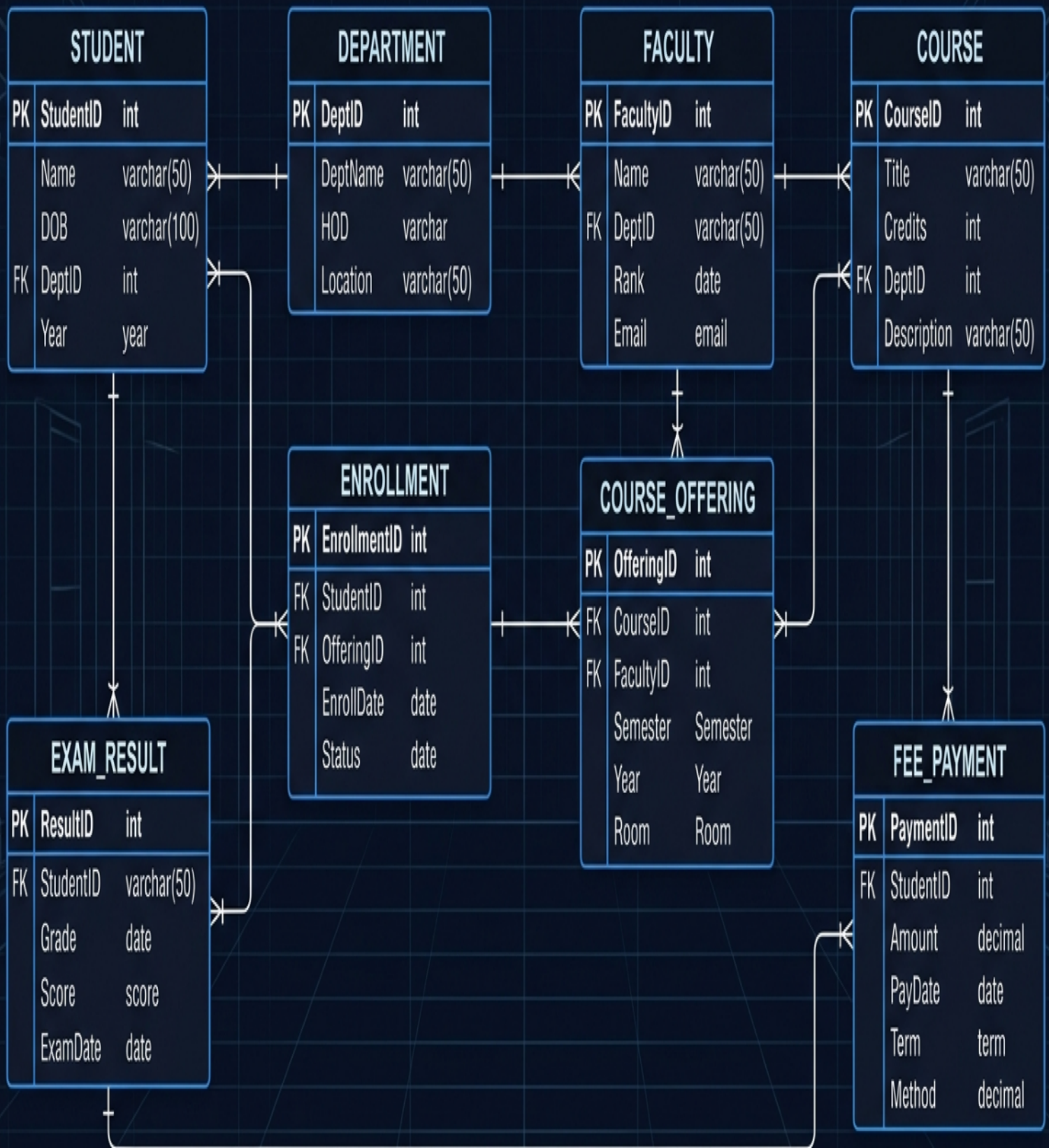
7. EXAM_RESULT

result_id (PK), enrollment_id (FK -> ENROLLMENT, UQ), marks_obtained (CHECK 0..100), grade (CHECK 'S'..'F'), exam_date (DEFAULT)

8. FEE_PAYMENT

payment_id (PK), student_id (FK -> STUDENT), amount_paid (CHECK > 0), payment_date (DEFAULT), payment_method (CHECK)

College Management System - BCSE302L Database Systems | Entity-Relationship (ER) Diagram



4. VIVA DEFENSE & FACULTY SIGN-OFF

BCSE302L - REVIEW I

Q: Why separate COURSE from COURSE_OFFERING?

A: A catalog course represents an invariant syllabus item. A course offering represents a specific term section (venue, t

Q: How do you enforce class size limits?

A: Via COURSE_OFFERING(max_capacity, current_enrolled) with CHECK constraints, plus PL/SQL procedure Enroll_Stude

Q: Why is EXAM_RESULT mapped 1:1 to ENROLLMENT?

A: A student must be officially registered to take an exam. Mapping to ENROLLMENT with a UNIQUE constraint guarante

FACULTY EVALUATION RUBRIC (REVIEW I)

- Problem Definition & Institutional Scope (20%): [/ 20]
- Entity-Relationship Diagram & Cardinalities (30%): [/ 30]
- Relational Schema & BCNF Normalization (30%): [/ 30]
- Viva Defense & Technical Justification (20%): [/ 20]

Student Team Signature

Faculty Signature (Prof. Ramkumar T)